

THE CONE IN A BATH: BOSON, GRAVITATIONAL, AND PLASMA RELAXATION OF THE SEMI-MAJOR AXIS

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ABSTRACT. The Schwinger-boson dictionary of [5] identifies the cone coordinate T with the total occupation number of two oscillator modes, $T = (n_1 + n_2 + 1)/2$, and the perturbed Levi-Civita regularization of [6] gives an exact classical equation for $\dot{T}$ under an arbitrary perturbing acceleration. This note asks what happens to T when the perturbation is not a single deterministic force but a genuine bath. On the quantum side, coupling each oscillator independently to a thermal bath at mean occupation $\bar{n}$ gives *exact* linear relaxation, $\dot{T} = -\kappa(T - \bar{n} - \frac{1}{2})$ and $\dot{X} = -\kappa(X - \frac{1}{2})$, with the third cone coordinate satisfying $\langle Y^2 \rangle_{ss} = \bar{n}(\bar{n} + 1)$ at steady state — consistent with the cone identity itself, $\langle X \rangle^2 + \langle Y \rangle^2 = \langle T \rangle^2$, which we verify holds exactly. On the classical side, modeling a gravitational bath as a stochastic velocity process with drift $\mathbf{A}(\mathbf{v})$ and diffusion tensor $D(\mathbf{v})$, a direct application of Itô's lemma gives $\langle \dot{T} \rangle = \frac{2T^2}{\mu} (\mathbf{v} \cdot \mathbf{A} + \text{Tr } D) + \frac{8T^3}{\mu^2} \mathbf{v}^\top D \mathbf{v}$: an extra, always-nonnegative term beyond the naive chain-rule guess, present because $T = -\mu/(2h)$ is a convex function of energy. We report the numerical process by which the missing term was caught (a 20σ disagreement against a 20,000-trajectory ensemble simulation) and verify the corrected formula to 0.09σ . Since the underlying theorem holds for any drift and diffusion, we derive both the diffusion coefficient and (from the same binary-encounter framework) the dynamical-friction coefficient for a charged particle in a plasma directly from Rutherford scattering, verify each against 3×10^6 simulated encounters to the $O(1/\ln \Lambda)$ accuracy expected of a leading-log calculation, and find that for equal test/field masses the two combine to cancel exactly, isolating the pure Itô/Jensen heating term for this idealized (monoenergetic, non-Maxwellian) bath. Both derivations caught and corrected their own arithmetic errors along the way (a too-small impact-parameter cutoff in one case, a mis-simplified power of the relative speed in the other), each caught by comparison against the numerical data. We discuss why the Itô correction has no counterpart on the quantum side — T is exactly linear in the boson numbers, but only reciprocally related to the classical conserved energy — and we are explicit about the substantial classical astrophysics and plasma physics (Chandrasekhar dynamical friction, two-body relaxation, Coulomb collision theory, and the absence of a true global equilibrium for self-gravitating systems) that this note only partially imports.

1. TWO BATHS, ONE COORDINATE

[5] shows $T - \frac{1}{2} = j = (n_1 + n_2)/2$: the cone's own semi-major-axis coordinate [1] is, up to an additive constant, half the total number of quanta in the two-oscillator Schwinger realization. [6] separately derives an exact classical equation, $\dot{T} = \frac{2T^2}{\mu} (\mathbf{P} \cdot \mathbf{v})$, for how T responds to an arbitrary perturbing acceleration $\mathbf{P}$. Both results concern a *single*, deterministic perturbation. This note asks the natural next question on each side: what happens to T under a genuine *bath* — many independent, statistically-specified interactions — rather than one prescribed force or drive.

2. THE BOSON BATH

Couple each oscillator independently to a thermal reservoir at mean occupation $\bar{n}$, with damping rate κ , via the standard two-mode Lindblad master equation

$$\dot{\rho} = -i[H, \rho] + \sum_{i=1,2} \left[\kappa(\bar{n} + 1) \mathcal{D}[a_i] \rho + \kappa \bar{n} \mathcal{D}[a_i^\dagger] \rho \right], \quad \mathcal{D}[L] \rho := L \rho L^\dagger - \frac{1}{2} \{L^\dagger L, \rho\}.$$

Theorem 2.1 (Exact linear relaxation of T and X). *Under the master equation above,*

$$(1) \quad \dot{T} = -\kappa \left(T - \bar{n} - \frac{1}{2} \right), \quad \dot{X} = -\kappa \left(X - \frac{1}{2} \right).$$

Proof. For a single mode with this dissipator, $d\langle n \rangle / dt = -\kappa(\langle n \rangle - \bar{n})$ is the standard input-output result [7]; applying it to each independent mode and using $T = (n_1 + n_2 + 1)/2$, $X = (n_1 - n_2 + 1)/2$ gives (1) directly, with $\bar{n}$ cancelling in the difference defining X . $\square$

Numerical Verification. Simulating the single-mode master equation directly on a truncated ($N = 40$) Fock space from the non-thermal initial state $|5\rangle\langle 5|$, with $\kappa = 0.7$, $\bar{n} = 2.3$, the resulting $\langle n(t) \rangle$ matches the analytic solution $\bar{n} + (n_0 - \bar{n})e^{-\kappa t}$ to within 1.9×10^{-3} over $t \in [0, 8]$ (RK4, $dt = 10^{-3}$), and $\text{tr } \rho$ remains 1 to machine precision throughout.

Proposition 2.2 (Steady-state third coordinate). *Y is not itself a well-defined observable, but $Y^2 = xy = J_- J_+ = a_2^\dagger a_1 a_1^\dagger a_2$ is Hermitian. At the thermal steady state (both modes independently thermal at $\bar{n}$),*

$$(2) \quad \langle Y^2 \rangle_{\text{ss}} = \bar{n}(\bar{n} + 1),$$

and the cone identity survives exactly at the level of these mixed moments: $\langle X \rangle^2 + \langle Y^2 \rangle_{\text{ss}} = \langle T \rangle^2$.

Proof. At steady state the two modes factorize, so $\langle Y^2 \rangle_{\text{ss}} = \langle a_1 a_1^\dagger \rangle \langle a_2^\dagger a_2 \rangle = (\bar{n} + 1)\bar{n}$. For the identity: $\langle X \rangle^2 + \langle Y^2 \rangle_{\text{ss}} = \frac{1}{4} + \bar{n}^2 + \bar{n} = (\bar{n} + \frac{1}{2})^2 = \langle T \rangle^2$. $\square$

Numerical Verification. Computing the exact thermal-state moments on a truncated ($N = 60$) Fock space at $\bar{n} = 2.3$: $\langle Y^2 \rangle = 7.58999987$ against $\bar{n}(\bar{n} + 1) = 7.59$ (agreement limited by truncation), and $\langle X \rangle^2 + \langle Y^2 \rangle - \langle T \rangle^2 = -1.8 \times 10^{-15}$.

Remark (What is not closed). Proposition 2.2 is a steady-state statement only. Away from steady state, $\langle Y^2(t) \rangle$ involves a genuine fourth moment of the ladder operators, which closes into a small system of ODEs only for Gaussian states (via Wick's theorem); a sharp Fock state such as $|n_1, n_2\rangle$ is not Gaussian, and the bath drives it through non-Gaussian intermediate states on its way to the (Gaussian) thermal steady state. We have not derived a transient equation for $\langle Y^2(t) \rangle$ and do not claim one is available in closed form for arbitrary initial states.

3. THE GRAVITATIONAL BATH

Model the effect of many weak gravitational encounters on the physical (unregularized) velocity as a Langevin process, $d\mathbf{v} = \mathbf{A}(\mathbf{v}) dt + B dW$, with diffusion tensor $D := \frac{1}{2} B B^\top$, superposed on the two-body acceleration. Let $h = \frac{1}{2} |\mathbf{v}|^2 - \mu/r$ and $T = -\mu/(2h)$ as before.

Theorem 3.1 (Itô-corrected drift of T).

$$(3) \quad \langle \dot{T} \rangle = \frac{2T^2}{\mu} (\mathbf{v} \cdot \mathbf{A} + \text{Tr } D) + \frac{8T^3}{\mu^2} \mathbf{v}^\top D \mathbf{v}.$$

Both correction terms are always ≥ 0 for any physical (positive semi-definite) D .

Proof. By Itô's lemma applied directly to $h = \frac{1}{2}|\mathbf{v}|^2 - \mu/r$ under the velocity process (the potential term contributes no stochastic part, since only $\mathbf{v}$ is driven by noise): $dh = \mathbf{v} \cdot d\mathbf{v} + \frac{1}{2} d\mathbf{v} \cdot d\mathbf{v} = [\mathbf{v} \cdot \mathbf{A} + \text{Tr } D] dt + \mathbf{v}^\top B dW$, using $\langle d\mathbf{v} d\mathbf{v}^\top \rangle = 2D dt$. So h itself is a diffusion with drift $\mathbf{v} \cdot \mathbf{A} + \text{Tr } D$ and instantaneous variance rate $\sigma_h^2 = \mathbf{v}^\top (2D) \mathbf{v}$. Applying Itô's lemma once more to $T(h) = -\mu/(2h)$, using $T'(h) = 2T^2/\mu$ and $T''(h) = 8T^3/\mu^2$:

$$\langle \dot{T} \rangle = T'(h) [\mathbf{v} \cdot \mathbf{A} + \text{Tr } D] + \frac{1}{2} T''(h) \sigma_h^2 = \frac{2T^2}{\mu} (\mathbf{v} \cdot \mathbf{A} + \text{Tr } D) + \frac{8T^3}{\mu^2} \mathbf{v}^\top D \mathbf{v}. \quad \square$$

How the second term was found. A first version of this note claimed only the first term of (3), obtained by differentiating T with respect to the *mean* of h and substituting the mean drift — the natural but incomplete generalization of [6]'s deterministic Corollary. Simulating an ensemble of 20,000 independent noisy trajectories (isotropic constant D , linear friction $\mathbf{A} = -\gamma\mathbf{v}$, μ, T_0, e_0 as in [2], $t \in [0, 40\text{s}]$) and comparing the ensemble mean $\langle T(t_{\text{end}}) \rangle$ against that first formula gave a 20σ discrepancy, always in the same direction (the simulation ran hotter). Since T is convex in h , this is exactly the signature of a missing Jensen/Itô second-derivative term; deriving it, as above, and re-running the same ensemble gave agreement to 0.086σ (Table 1). We record the initial error and its diagnosis deliberately: it is the numerical verification itself that is the load-bearing part of this note's method, not any individual formula.

	T at $t = 40\text{s}$	
Monte Carlo ensemble ($n = 20,000$)	25105.51 ± 30.25	
Formula without Itô correction	24494.04	(20.2 σ off)
Formula (3), with correction	25108.10	(0.09 σ off)

Table 1. Parameters: $\mu = 398600.4418 \text{ km}^3/\text{s}^2$, $T_0 = 24482 \text{ km}$, $e_0 = 0.72$, $\gamma = 3 \times 10^{-6} \text{ s}^{-1}$, isotropic $D = 2 \times 10^{-4} \text{ km}^2/\text{s}^3$, Euler-Maruyama, $dt = 0.01\text{s}$.

Remark (Scope: what this is not). This note uses a toy, state-independent (or simply velocity-linear) D for the numerical check. The classical theory of a real gravitational bath — Chandrasekhar dynamical friction and two-body relaxation [8, 9] — gives $\mathbf{A}(\mathbf{v})$ and $D(\mathbf{v})$ specific, velocity-dependent forms set by the background mass density ρ , velocity dispersion σ , and Coulomb logarithm $\ln \Lambda$; Theorem 3.1 holds for any such $\mathbf{A}, D$, but we have not substituted the Chandrasekhar forms or connected $\kappa, \bar{n}$ -language to ρ, σ for an astrophysical bath. Nor do we claim a global equilibrium exists for a genuinely self-gravitating system: negative specific heat and the gravothermal catastrophe are well known obstructions to that [9], and the physics here — like Chandrasekhar's own — only concerns one test orbit relaxing toward local equipartition with a fixed background, not the thermalization of the whole system.

4. THE ELECTRIC (PLASMA) BATH

Theorem 3.1 holds for *any* $\mathbf{A}(\mathbf{v}), D(\mathbf{v})$; nothing in its proof is specific to gravity. For a test charge moving through a plasma — density n_f , background charge q_f , test charge

q_t , reduced mass μ_{red} — the same Coulomb $1/r^2$ scattering that underlies Chandrasekhar's formula gives the diffusion tensor directly, and here we derive it from individual binary (Rutherford) encounters rather than quote it, since the numerical-verification standard of this note requires it.

Proposition 4.1 (Diffusion coefficient from binary Coulomb encounters). *For a test particle of mass m_t moving at relative speed u through field particles of density n_f , coupling $\kappa_{\text{int}} = q_t q_f / (4\pi\epsilon_0)$, the small-angle (leading-log) perpendicular diffusion coefficient is*

$$(4) \quad \text{Tr } D = \frac{8\pi \kappa_{\text{int}}^2 n_f \ln \Lambda}{m_t^2 u}, \quad \ln \Lambda = \ln(b_{\text{max}}/b_{90}), \quad b_{90} = \frac{\kappa_{\text{int}}}{\mu_{\text{red}} u^2}.$$

Proof. For an encounter at impact parameter b , the exact Rutherford deflection is $\theta = 2 \arctan(b_{90}/b)$; for $b \gg b_{90}$, $\theta \approx 2b_{90}/b$, and the resulting lab-frame kick is $|\Delta v| \approx \frac{m_f}{m_t + m_f} u \theta = \frac{2\kappa_{\text{int}}}{m_t u b}$ (using $\mu_{\text{red}} = m_t m_f / (m_t + m_f)$). Field particles cross impact parameter $[b, b + db]$ at rate $n_f u \cdot 2\pi b db$, so

$$\text{Tr } D = \int_{b_{90}}^{b_{\text{max}}} \left(\frac{2\kappa_{\text{int}}}{m_t u b} \right)^2 n_f u 2\pi b db = \frac{8\pi \kappa_{\text{int}}^2 n_f}{m_t^2 u} \int_{b_{90}}^{b_{\text{max}}} \frac{db}{b} = \frac{8\pi \kappa_{\text{int}}^2 n_f \ln \Lambda}{m_t^2 u}. \quad \square$$

The lower cutoff is taken at b_{90} , the standard choice: below it the deflection is $O(1)$, not small, and the integrand above is no longer valid.

Numerical Verification. We simulated 3×10^6 trial time-windows, each containing a Poisson-distributed number of *exact* (not small-angle) Rutherford encounters at correctly-weighted impact parameters $b \in [b_{90}, 50b_{90}]$ ($\ln \Lambda = 3.91$), and measured $\text{Tr } D$ directly from the resulting velocity-kick statistics of a test particle initially at rest in an isotropic field of speed- u_0 scatterers ($\kappa_{\text{int}} = m_t = m_f = 1$, $u_0 = 2$, $n_f = 0.05$, arbitrary units). Result: $\text{Tr } D_{\text{MC}} = 2.241 \pm 0.002$ against the analytic prediction (4) of 2.458 — a 9% gap, entirely consistent with the well known fact that the coefficient inside $\ln \Lambda$ is only defined up to $O(1)$ (the precise value of b_{min} is a convention), so leading-log accuracy at $\ln \Lambda \approx 4$ is expected to be good only to $O(1/\ln \Lambda) \approx 25\%$. We initially used $b_{\text{min}} = 0.01 b_{90}$ and found the prediction off by a factor of 2.3; the fix (restoring $b_{\text{min}} = b_{90}$, the standard cutoff, since small-angle theory is invalid below it) resolved this, and we record the mistake for the same reason as Section 3's.

Proposition 4.2 (Dynamical friction from the same binary encounters). *Under the same small-angle, leading-log approximation as Proposition 4.1, the longitudinal (friction) drift is*

$$(5) \quad \langle \Delta v_{\parallel} \rangle / \Delta t = - \frac{4\pi \kappa_{\text{int}}^2 (m_t + m_f) n_f \ln \Lambda}{m_t^2 m_f u^2},$$

and consequently

$$(6) \quad \frac{\langle \Delta v_{\parallel} \rangle / \Delta t}{\text{Tr } D} = - \frac{m_t + m_f}{2m_f u} \xrightarrow{m_t = m_f} - \frac{1}{u}.$$

Proof. For an encounter at impact parameter $b \gg b_{90}$, the exact elastic parallel kick is $\Delta v_{\parallel} = \frac{m_f}{m_t + m_f} u (\cos \theta - 1)$, small-angle-expanded to $-\frac{m_f}{m_t + m_f} u \theta^2 / 2 = -\frac{2m_f \kappa_{\text{int}}^2}{(m_t + m_f) \mu_{\text{red}}^2 u^3 b^2}$. Integrating against the encounter rate $n_f u \cdot 2\pi b db$ from b_{90} to b_{max} and simplifying with $\mu_{\text{red}} = m_t m_f / (m_t + m_f)$ gives (5); dividing by (4) gives (6), verified symbolically. $\square$

Numerical Verification. Tracking the exact (non-small-angle) parallel component of the same 3×10^6 -encounter-window simulation as Proposition 4.1 gives $\langle \Delta v_{\parallel} \rangle / \Delta t = -1.120 \pm 0.001$ against the analytic prediction (5) of -1.229 — a 9% gap, the same size and same leading-log origin as Proposition 4.1’s. This calculation caught its own bookkeeping error along the way: an initial version of (5) had the wrong power of u (a stray factor of $1/u$ from mis-simplifying the algebra by hand), giving a value $2 \times$ too small; comparing directly against a symbolic recomputation, and against the same Monte Carlo data, caught it immediately, since (6) is a clean closed form and the erroneous version visibly violated it.

Corollary 4.3 (Semi-major-axis diffusion under a plasma bath, completed). *For equal test/field masses, combining (4)–(6) with Theorem 3.1 (using $\mathbf{v} \cdot \mathbf{A} = u \cdot \langle \Delta v_{\parallel} \rangle / \Delta t = -\text{Tr } D$ for this single-speed-shell model) gives*

$$(7) \quad \langle \dot{T} \rangle = \frac{2T^2}{\mu} (\text{Tr } D - \text{Tr } D) + \frac{8T^3}{\mu^2} \mathbf{v}^{\top} D \mathbf{v} = \frac{8T^3}{\mu^2} \mathbf{v}^{\top} D \mathbf{v} \geq 0.$$

Remark. Equation (7) says that for a test particle scattering off a monoenergetic (single-speed, not Maxwellian) field of identical particles, the leading friction and diffusion contributions to $\langle \dot{T} \rangle$ cancel *exactly*, leaving only the strictly positive Itô/Jensen term — the same convexity heating identified in Section 3, here isolated with no competing loss term at all. This is a feature of the idealized monoenergetic-shell model, not a general plasma-physics result: a genuine Maxwellian bath at temperature Θ has a nonzero net drift tying $\langle T \rangle$ to Θ , which requires averaging (4)–(6) over the field’s speed distribution rather than evaluating them at a single u , and we have not done that here.

Remark. Unlike the astrophysical case, this bath is directly testable: electron-ion relaxation rates in laboratory and fusion plasmas are routinely measured, so (4)–(6), combined with a proper Maxwellian average, would in principle be checkable against real data rather than only against simulation. We have not made that comparison.

5. DISCUSSION: WHY THE CORRECTION IS ONE-SIDED

Table 2 summarizes both results. The Itô/Jensen correction appears only on the classical side, and the reason is structural rather than accidental: in the quantum dictionary T is *linear* in the additively-conserved quantity the bath actually exchanges, the boson number $N = n_1 + n_2$; in the classical regularized picture, the quantity the bath exchanges is energy h , and $T = -\mu/(2h)$ is only a *nonlinear* (reciprocal) function of it. Itô corrections arise exactly where the coordinate of interest is a nonlinear function of the variable actually undergoing additive diffusion — so the asymmetry between the two bath pictures is a statement about which coordinate is linearly exchanged in each formalism, not a difference in the underlying physics of dissipation and fluctuation.

	boson bath	gravitational bath	electric bath (plasma)
exchanged quantity	$N = n_1 + n_2$ (linear in T)	h (reciprocal in T)	h (reciprocal in T)
loss term	$-\kappa T$	$\frac{2T^2}{\mu} \mathbf{v} \cdot \mathbf{A}$	$-\frac{2T^2}{\mu} \text{Tr } D$ (derived, eq. 6; exactly cancels gain for equal masses)
gain term	$+\kappa(\bar{n} + \frac{1}{2})$	$\frac{2T^2}{\mu} \text{Tr } D$	$\frac{2T^2}{\mu} \cdot \frac{8\pi\kappa_{\text{int}}^2 n_f \ln \Lambda}{m_i^2 u}$ (derived, eq. 4)
extra (Jensen) term	none (T linear)	$\frac{8T^3}{\mu^2} \mathbf{v}^\top D \mathbf{v}$	same term; isolated exactly (eq. 7) for the monoenergetic-shell model
equilibrium	$T_{\text{eq}} = \bar{n} + \frac{1}{2}$	requires $\mathbf{A}, D$ balance; not derived	Maxwellian average not done; shell model has no drift term at all
testable directly?	in principle (cavity QED)	no (astrophysical)	yes (lab/fusion plasmas)

Table 2.

AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic’s Claude) were used in the derivation, symbolic and numerical cross-checking, and L^AT_EX preparation of this note. This includes: numerical verification of Theorem 2.1 against direct master-equation simulation; numerical verification of Proposition 2.2 against exact thermal-state moments; symbolic (Taylor-series) verification of the Itô expansion underlying Theorem 3.1; the 20,000-trajectory Monte Carlo ensemble verification of Theorem 3.1, including the initial 20σ discrepancy against an incomplete first version of the formula, its diagnosis as a missing convexity/Itô term, and confirmation of the corrected formula to 0.09σ ; and the 3×10^6 -trial binary-encounter Monte Carlo verification of Propositions 4.1 and 4.2, including an initial factor-of-2.3 discrepancy in the diffusion coefficient (traced to an incorrectly small b_{min} cutoff) and a separate factor-of-2 discrepancy in the friction coefficient (traced to a mis-simplified power of u), both caught by comparison against the numerical data and corrected. All mathematical claims were independently verified by the author.

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